



Petty Cash Monitoring System

System Proposal

Prepared for: L.V. Ledesma Construction, Inc.

Project Location: Uptown Cagayan De Oro

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1. Executive Summary

This proposal presents the design and development of a **Petty Cash Monitoring System** for **L.V. Ledesma Construction, Inc.** The system addresses the challenges of manually tracking petty cash funds, expenses, and liquidation processes by providing a centralized, web-based platform with automated monitoring, threshold alerts, and formal report generation.

The system maintains a fixed petty cash fund of **₱30,000**, automatically tracks expenses against the fund balance, triggers replenishment alerts at 30% utilization, and generates formal liquidation reports exportable to Excel. It is built using **Laravel 13** (PHP), **PostgreSQL**, and is deployed on **Render** for cloud accessibility.

2. Problem Statement

Construction companies like L.V. Ledesma Construction, Inc. commonly manage petty cash funds for day-to-day operational expenses such as supplies, transportation, meals, and communication. The current manual process presents several challenges:

- **Lack of Real-Time Visibility:** Fund balances and expense totals are not easily trackable, leading to overspending or delayed replenishment.
- **Manual Record-Keeping:** Paper-based or spreadsheet tracking is prone to errors, data loss, and duplication.
- **Delayed Liquidation:** Without automated alerts, liquidation reports are often prepared late, causing discrepancies in fund reconciliation.
- **No Audit Trail:** Manual systems lack proper logging of who created, modified, or approved transactions.
- **Inefficient Replenishment:** The process of requesting, approving, and disbursing replenishment funds is slow and lacks transparency.

3. Proposed Solution

A **web-based Petty Cash Monitoring System** that provides:

- Real-time fund balance tracking with a fixed ₱30,000 target
- Automated expense recording with balance deduction
- 30% threshold auto-trigger for replenishment requests
- Multi-step replenishment workflow (Request → Approve → Disburse)
- Formal liquidation report generation with grouped expense items

- Excel export and print-ready report formatting
- Role-based access control (Admin / User)

4. Objectives

4.1 General Objective

To develop a web-based petty cash monitoring system that automates fund tracking, expense management, replenishment workflows, and liquidation reporting for L.V. Ledesma Construction, Inc.

4.2 Specific Objectives

1. Implement a centralized fund management system with a fixed ₱30,000 petty cash target.
2. Enable real-time expense recording with automatic fund balance updates.
3. Provide automated 30% threshold alerts that trigger replenishment report creation.
4. Develop a multi-step replenishment workflow with approval and disbursement tracking.
5. Generate formal liquidation reports with grouped expense items and signature blocks.
6. Support Excel export and print formatting for physical filing requirements.
7. Implement role-based access control to ensure proper authorization.
8. Maintain a complete audit trail of all financial transactions.

5. Scope and Limitations

Scope

- Petty cash fund creation, replenishment, and balance management
- Expense recording with payee, category, cost code, particular, amount, and date
- Automated 30% threshold detection and replenishment request generation
- Replenishment request workflow: Create → Approve/Reject → Disburse
- Liquidation report creation with auto-populated expense items
- Expense status tracking (Open/Closed) linked to liquidation reports
- Excel export of liquidation reports
- Print-ready report formatting
- User authentication and role-based access (Admin, User)

Limitations

- Single-fund system (one ₱30,000 fund at a time)
- No multi-currency support
- No email/notification system (alerts are in-app only)

- No mobile app (responsive web only)
- No integration with accounting software
- No receipt image upload

6. System Features and Functionality

6.1 Authentication and User Roles

Feature	Admin	User
View Dashboard	Yes	Yes
Create/Edit/Delete Funds	Yes	No
Create Expenses	Yes	Yes
Edit/Delete Expenses	Yes	No
Create Replenishment Requests	Yes	Yes
Approve/Reject/Disburse Replenishments	Yes	No
Create/Edit/Delete Reports	Yes	No
View Reports	Yes	Yes
Export/Print Reports	Yes	Yes

6.2 Dashboard

The dashboard provides a real-time overview including:

- Four key metric cards: Total Funds, Total Balance, Total Spent, Pending Alerts
- Pending replenishment requests table with Approve/Reject actions
- Petty Cash Funds table with utilization progress bars and status badges
- Recent Transactions table (10 most recent expenses)
- Add Fund (Replenish) modal for quick fund additions

6.3 Fund Management

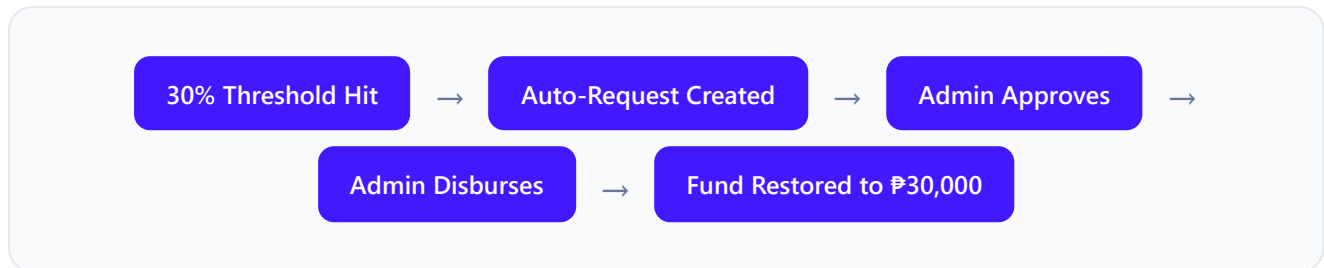
- **Target Balance:** Fixed at ₱30,000
- **Add Fund:** Increases current balance (replenishment)
- **Utilization Tracking:** Progress bar with color-coded thresholds (green <30%, yellow 30-50%, red >50%)
- **Status Tracking:** Active, To Liquidate, Pending

6.4 Expense Tracking

- Fields: Payee, Category, Particular, Cost Code, Amount, Receipt #, Date

- Real-time fund balance deduction upon expense creation
- Balance Preview on edit showing +/- difference and projected fund balance
- Expense status: Open (pending liquidation) / Closed (included in report)

6.5 Replenishment Workflow



- Auto-triggered when total expenses $\geq$ ₱9,000 (30% of ₱30,000)
- Manual creation available via Replenishments page
- Status flow: Pending $\rightarrow$ Approved $\rightarrow$ Disbursed (or Rejected)
- Disbursement restores fund balance to ₱30,000

6.6 Liquidation Reports

- Auto-populated with all open expenses
- Cash Received auto-filled with total liquidation amount
- Items grouped by Cost Code and Particulars in the report view
- For Return / (Reimbursement) auto-calculated
- After creation, included expenses are marked as "Closed"
- Signature blocks for Prepared By, Reviewed By, Verified By

6.7 Excel Export and Printing

- Download as `.xls` file with company header, itemized table, totals, and signatures
- Print-ready formatting optimized for A4 paper
- Navigation bar hidden in print view

7. System Architecture

7.1 Technology Stack

Layer	Technology
Backend Framework	Laravel 13.21 (PHP 8.3)
Database	PostgreSQL (Render Managed)
Frontend	Blade Templates, HTML5, CSS3, Vanilla JavaScript
Authentication	Laravel Breeze / Built-in Auth
Hosting	Render (PaaS)
Version Control	GitHub
Font	Inter (Google Fonts)

7.2 Database Schema

Core tables and their relationships:

users

Column	Type	Description
id	BIGINT (PK)	User identifier
name	VARCHAR(255)	Full name
email	VARCHAR(255)	Email (unique)
password	VARCHAR(255)	Hashed password
role	VARCHAR(20)	"admin" or "user"

petty_cash_funds

Column	Type	Description
id	BIGINT (PK)	Fund identifier
total_amount	DECIMAL(10,2)	Target allocation (₱30,000)
current_balance	DECIMAL(10,2)	Available balance

status	VARCHAR(30)	active / low_balance / replenishment_pending
timestamps	TIMESTAMP	created_at, updated_at

expenses

Column	Type	Description
id	BIGINT (PK)	Expense identifier
fund_id	BIGINT (FK)	References petty_cash_funds.id
payee	VARCHAR(255)	Payee name
category	VARCHAR(100)	Expense category
particular	VARCHAR(255)	Description (nullable)
cost_code	VARCHAR(100)	Department/project code
amount	DECIMAL(10,2)	Expense amount
receipt_number	VARCHAR(100)	Receipt reference (nullable)
expense_date	DATE	Date of expense
status	VARCHAR(20)	open / closed
timestamps	TIMESTAMP	created_at, updated_at

replenishment_requests

Column	Type	Description
id	BIGINT (PK)	Request identifier
fund_id	BIGINT (FK)	References petty_cash_funds.id
requested_amount	DECIMAL(10,2)	Amount requested
status	VARCHAR(20)	pending / approved / disbursed / rejected
triggered_by	TEXT	Auto-trigger description or "Manual Request"
timestamps	TIMESTAMP	created_at, updated_at

replenishment_reports

Column	Type	Description
id	BIGINT (PK)	Report identifier

project_name	VARCHAR(255)	Project name
location	VARCHAR(255)	Project location
subject	VARCHAR(255)	Report subject
period_start	DATE	Reporting period start
period_end	DATE	Reporting period end
report_date	DATE	Date report was created
cash_received	DECIMAL(10,2)	Initial cash amount
prepared_by	VARCHAR(255)	Prepared by name
reviewed_by	VARCHAR(255)	Reviewed by name
verified_by	VARCHAR(255)	Verified by name
timestamps	TIMESTAMP	created_at, updated_at

replenishment_items

Column	Type	Description
id	BIGINT (PK)	Item identifier
replenishment_report_id	BIGINT (FK)	References replenishment_reports.id
expense_id	BIGINT (FK)	References expenses.id (nullable)
expense_date	DATE	Expense date
voucher_no	VARCHAR(100)	Voucher number
reference_no	VARCHAR(100)	Reference number (nullable)
payee	VARCHAR(255)	Payee name
cost_code	VARCHAR(100)	Cost code
particulars	VARCHAR(255)	Particulars/description
amount	DECIMAL(10,2)	Item amount
group_key	TEXT	Grouping key for display
timestamps	TIMESTAMP	created_at, updated_at

7.3 Deployment Architecture



- **Source Control:** GitHub repository at github.com/joeyalod23/petty-cash-monitoring
- **CI/CD:** Automatic deployment on push to master branch
- **Database:** Render-managed PostgreSQL
- **URL:** <https://petty-cash-monitoring.onrender.com>

8. System Workflow

8.1 Complete Petty Cash Cycle



Step-by-Step Process:

- 1. Fund Initialization:** Admin creates/replenishes the petty cash fund to ₱30,000.
- 2. Expense Recording:** As expenses are incurred, they are logged into the system with payee, category, cost code, amount, and date. The fund balance is automatically reduced.
- 3. Threshold Monitoring:** The system continuously monitors total expenses. When they reach ₱9,000 (30% of ₱30,000):
 - A replenishment request is auto-generated
 - The user is redirected to create a liquidation report
- 4. Liquidation Report:** Admin creates a formal report with all open expenses, voucher numbers, and signature blocks. Expenses are marked as "Closed" after report creation.
- 5. Replenishment Processing:** Admin approves and disburses the replenishment request, restoring the fund to ₱30,000.
- 6. Cycle Reset:** The fund is restored and a new expense cycle begins.

9. User Interface Design

The system features a modern, professional interface with:

- **Color Scheme:** Dark navy (#1b2559) primary with purple (#4318ff) accent
- **Typography:** Inter font family with monospace for financial figures
- **Layout:** Sidebar navigation with main content area
- **Components:** Cards, data tables, progress bars, status badges, modals, alerts
- **Responsive:** Adapts to desktop, tablet, and mobile screens
- **Dark/Light:** Light theme with high-contrast elements

- **Real-time Clock:** Displayed in the top navigation bar

10. Security Considerations

- **Authentication:** Laravel's built-in session-based authentication with secure password hashing (bcrypt)
- **Authorization:** Role-based middleware (AdminMiddleware) restricts sensitive operations
- **CSRF Protection:** All forms include CSRF tokens
- **Input Validation:** Server-side validation on all form inputs
- **SQL Injection Prevention:** Eloquent ORM and parameterized queries
- **XSS Prevention:** Blade template auto-escaping of output
- **Database Transactions:** Critical operations wrapped in DB transactions for data integrity
- **Row Locking:** SELECT ... FOR UPDATE on fund balance modifications to prevent race conditions
- **Session Security:** Session regeneration on login, secure cookie handling
- **Environment Variables:** Sensitive configuration stored in `.env` (not in version control)

11. Project Timeline

Phase	Activity	Duration
1	Requirements Analysis & System Design	Week 1
2	Database Design & Schema Creation	Week 1
3	Backend Development (Models, Controllers, Services)	Week 2-3
4	Frontend Development (Views, UI/UX)	Week 2-3
5	Testing & Bug Fixing	Week 3-4
6	Deployment & Documentation	Week 4

12. Cost Analysis

Item	Cost	Notes
Render Hosting (Web Service)	Free Tier / \$7/mo	Automatic deployments from GitHub
Render PostgreSQL	Free Tier / \$7/mo	Managed database
Domain (optional)	~\$12/year	Custom domain mapping
Development	In-house	Internal development team

13. Conclusion and Recommendation

The proposed Petty Cash Monitoring System provides a comprehensive solution to the challenges of manual petty cash management. By implementing automated fund tracking, threshold alerts, formal liquidation reporting, and role-based access control, the system will significantly improve the efficiency, accuracy, and transparency of petty cash operations at L.V. Ledesma Construction, Inc.

Recommendations:

1. **Immediate Implementation:** Deploy the system for pilot testing with the current fund custodian.
2. **User Training:** Conduct training sessions for all staff who will interact with the system.
3. **Feedback Collection:** Gather user feedback after 30 days to identify areas for improvement.
4. **Future Enhancements:** Consider adding email notifications, receipt image uploads, multi-fund support, and accounting software integration in future versions.